Scalability & Future Plan

Date	15 March 2026
Team ID	SWTID-2026-6912
Project Name	FitBuddy - AI Fitness Plan Generator
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local File Database (SQLite)	Restricts high concurrent write operations and multi-region deployment.	High
2	Synchronous Gemini API Calls	Slight latency during plan generation when API traffic spikes.	Medium
3	Single Server Instance	Lack of automated load balancing limits high volume user sessions.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles limited concurrent users on single deployment instance.	Deploy on containerized infrastructure (Docker / Kubernetes) with auto-scaling.
2	Data Storage	SQLite database for local user and plan storage.	Migrate to scalable PostgreSQL / AWS RDS with Redis caching layer.
3	Performance	Direct synchronous HTTP requests to Gemini API.	Implement asynchronous job queues (Celery/RabbitMQ) for fast user response.
4	Security	Basic authentication and session handling.	Upgrade to OAuth2 / JWT with HTTPS, role-based access control, and rate limiting.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Wearable Device Integration (Apple Health / Fitbit sync)	Q2 2026	Real-time tracking of burned calories and activity levels.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Computer Vision Exercise Form Correction via Camera	Q3 2026	Prevents injuries by monitoring posture during exercise execution.
Phase 3	Multilingual Support & Localized Meal Plans	Q4 2026	Expands accessibility across diverse global user regions.
Phase 4	AI Community Challenges & Leaderboards	Q1 2027	Increases long-term user retention through social engagement.